1. Competency #2: Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem

(Highly relevant)

- The core of this task is assessing whether common-sense inferences about an average are logically valid.
- Students must recognize that "an average of 315" is a summary statistic that does not determine individual values.
- Statement 2, in particular, tests whether students can spot an impossible scenario that violates the definition of average—directly aligning with judging the reasonableness of conclusions.

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2. Competency #8: Recognise and explain the extent and limits of mathematical concepts and solutions

(Highly relevant)

- The question fundamentally assesses understanding of the limitations of the arithmetic mean:
  - It reveals nothing about the distribution of data;
  - It does not guarantee that any observation equals the mean;
  - Extreme values (like 0 days) are permissible as long as the total sum matches.
- Students must grasp that the mean is not a typical value nor a required value in the dataset.

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3. Competency #7: Construct and communicate explanations and arguments in the context of the problem

(Moderately relevant)

- Although the task only requires "Yes/No" responses, correct answers depend on internal logical reasoning.
- In instructional settings, students should be able to justify why Statement 2 is impossible (total would exceed 3,150) and why Statement 3 is possible (compensation by other years).
- Thus, this competency is implicitly essential for deep understanding.

4. Competency #3: Identify the limits of the model used to solve a problem

(Moderately relevant)

- The "model" here is the use of a single average to summarize 10 years of usage.
- The three statements prompt students to reflect: an average alone cannot reveal year-to-year variation.
- This directly examines awareness of the simplification inherent in using averages.

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5. Competency #6: Use real-world knowledge to judge how to adjust or apply mathematical results

(Weakly relevant)

- Although set in a real-world context (holiday rental), the reasoning is purely mathematical.
- No external assumptions (e.g., "apartments can't be unused for a whole year") are needed or appropriate—the analysis relies solely on numerical logic.
- Hence, real-world knowledge plays little role.

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6. Competency #1: Interpret a mathematical result back into the real-world context

(Weakly relevant)

- This task does not involve computing a result and then interpreting it; instead, it asks students to evaluate proposed interpretations of given data.
- The focus is on critical evaluation, not translation of a computed output—making this less central than #2 and #8.

1. 能力 #2：识别数学结果或结论在问题语境下是否合理

（高度相关）

> “Identify whether a mathematical result or conclusion does, or does not, make sense given the context of a problem.”

- 本题核心正是判断三个关于“平均值”的常识性推论是否成立。
- 学生需意识到：“平均为 315”是一个汇总统计量，不能反推具体年份的精确值或极端分布。
- 特别是陈述 2，若认为“每年都 >315”可能，则违背了平均值的基本数学性质——这直接考查学生能否识别“不合理”的结论。

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2. 能力 #8：认识并说明数学概念及其解的范围与局限

（高度相关）

> “Recognise [demonstrate, interpret, explain] the extent and limits of mathematical concepts and mathematical solutions.”

- 题目本质上考察对“算术平均数”这一数学概念局限性的理解：
  - 平均值不揭示分布形态；
  - 不保证存在等于平均值的数据点；
  - 极值（如 0 天）在数学上是允许的，只要总和匹配。
- 学生必须明确：平均值 ≠ 典型值 ≠ 必然出现的值。

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3. 能力 #7：构建并交流基于问题语境的解释与论证

（中等相关）

> “Construct and communicate explanations and arguments in the context of the problem.”

- 虽然题目只要求勾选 Yes/No，但正确作答依赖于内在逻辑推理。
- 在教学或形成性评价中，学生应能解释为何陈述 2 不可能（总和会超）、陈述 3 可能（可补偿）。
- 因此，该能力是隐含的高阶要求。

4. 能力 #3：识别所用模型的局限性

（中等相关）

> “Identify the limits of the model used to solve a problem.”

- 此处的“模型”即用单一平均值概括 10 年使用情况。
- 题目通过三个陈述，引导学生反思：仅凭平均值无法推断个体年份的具体行为。
- 这正是对“平均值作为简化模型”的局限性的直接考察。

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5. 能力 #6：利用现实世界知识判断如何调整或应用数学结果

（弱相关）

> “Use knowledge of how the real world impacts the outcomes… to make contextual judgments…”

- 虽然涉及真实情境（度假公寓使用），但本题纯属数学逻辑推理，不依赖外部常识（如“公寓不可能全年空置”等）。
- 所有判断均可在数学内部完成，无需现实约束修正。
- 因此此项相关性较低。

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6. 能力 #1：将数学结果还原到现实语境

（弱相关）

> “Interpret a mathematical result back into the real world context.”

- 题目并非先计算再解释，而是直接评估对现实数据的推论是否成立。
- 更侧重“批判性解读”而非“结果回译”，故此项不如 #2 和 #8 核心。